

Exercise Sheet 15

Discrete Mathematics by Hongfei Fu, 2023.11.27

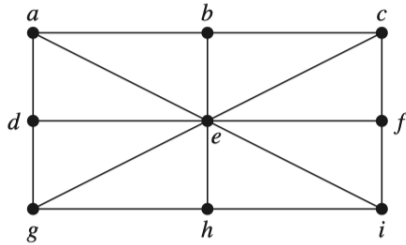
Exercise Sheet 15

Discrete Mathematics by Hongfei Fu, 2023.11.27

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1. ([R], Page 705, Exercise 36) In Exercises 36 determine whether the given graph has a Hamilton circuit. If it does, find such a circuit. If it does not, give an argument to show why no such circuit exists.

36.



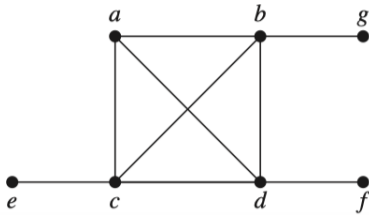
Answer Area:

Yes, there is a Hamilton circuit:

$$a, ad, d, dg, g, gh, h, hi, i, if, f, fc, c, ce, e, eb, b, ba$$

2. ([R], Page 705, Exercise 40) Does the graph in Exercise 33 have a Hamilton path? If so, find such a path. If it does not, give an argument to show why no such path exists.

33.



Answer Area:

No. There isn't a Hamilton path. A proof is as below:

Assuming that there is a Hamilton path $v_1, \dots, v_n$.

$\therefore$ Every vertex in the graph appears only once in the path.

∴ Except for v_1 and v_n , each path has an edge that arrive at it and an edge that leave from it. That is to say, each vertex's degree is at least 2 except for v_1 and v_n .

$\therefore$ There are three vertices in the graph whose degree is 1: e, f, g .

$\therefore$ There isn't a Hamilton path.

3. ([R], Page 706, Exercise 49, **2pt**) Show that there is a Gray code of order n whenever n is a positive integer, or equivalently, show that the n -cube Q_n , $n > 1$, always has a Hamilton circuit. [Hint: Use mathematical induction. Show how to produce a Gray code of order n from one of order $n - 1$.]

Answer Area:

- When $n = 1$, there is a Gray code of order n : 0, 1
- Suppose that there is a Gray code of order $n - 1$ ($n > 1$), then we can construct a Gray code of order n as follows:

let the Gray code of order $n - 1$ be $v_1, v_2, \dots, v_k$.

First, we add 0 in front of each number in the order:

$$0v_1, 0v_2, \dots, 0v_k$$

According to the definition of Gray code, each increment will only cause one-bit flip as the original one.

Then, we add 1 in front of each number in the original order, and reverse the order then add it to the newly constructed order above:

$$0v_1, 0v_2, \dots, 0v_n, 1v_n, v_{n-1}, \dots, 1v_1$$

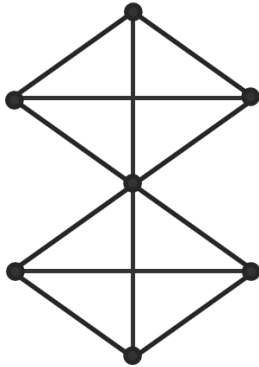
Therefore, we construct a Gray code of order n which contains 2^n numbers.

In conclusion, according to the mathematical induction, there is a Gray code of order n whenever n is a positive integer

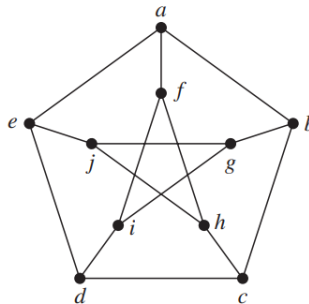
4. (Exercise 48, **2pt**) Can you find a simple connected graph with n vertices with $n \geq 3$ that does not have a Hamilton circuit, yet the degree of every vertex in the graph is at least $(n - 1)/2$?

Answer Area:

Yes:



5. (Exercise 46, **1pt**) Show that any subgraph of the Petersen graph below obtained by deleting



a vertex has a Hamilton circuit.

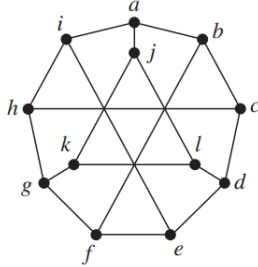
Answer Area:

We can divide the problem into two cases by symmetry:

- Delete a vertex in $\{a, b, c, d, e\}$. Assuming that we delete a.
There is a Hamilton circuit:
 $b, c, d, e, j, h, f, i, g, b$

- Delete a vertex in $\{f, g, h, i, j\}$. Assuming that we delete f .
There is a Hamilton circuit:
 $a, b, g, i, d, c, h, j, e, a$

6. (Exercise 67, **2pt**) Show that this graph does not have a Hamilton circuit



Answer Area:

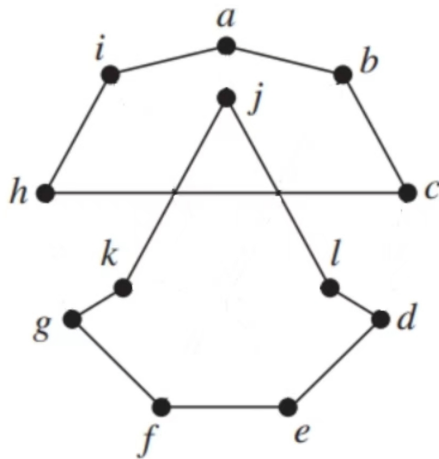
Assuming that there is a Hamilton circuit. Then every vertex has at least an edge and an edge out

$\therefore kj, jl, lk$ can't be in the circuit at the same time. (Otherwise when the circuit head for the vertices in the outside circle, it must pass a vertex in $\{k, j, l\}$ twice.)

$\therefore$ Let's assume that kl isn't in the circuit. Then kj, jl must be in the circuit and kg, ld must be in the circuit to connect with the vertices in the outside circle. And aj must not be in the circuit, otherwise there will be an edge arrive at j but not an edge leave it.

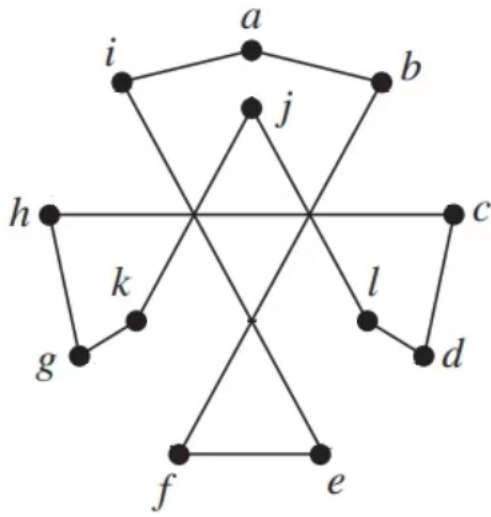
Then since $\deg(a) \geq 2$, ai, ab must be in the circuit.

- If ie not in the circuit.
 ie not in the circuit $\rightarrow ih$ in the circuit, ef, ed in the circuit
 $\rightarrow cd$ not in the circuit $\rightarrow cb, ch$ in the circuit
 $\rightarrow bf, hg$ not in the circuit $\rightarrow gf$ in the circuit.
 $\therefore$ If we delete the edges not in the circuit, the graph will be like this:



Apparently nor a circuit. Contradict.

- If ie in the circuit.
 ie in the circuit $\rightarrow ih$ not in the circuit, hc, hg in the circuit
 $\rightarrow gf$ not in the circuit $\rightarrow fb, fe$ in the circuit
 $\rightarrow bc, ed$ not in the circuit $\rightarrow cd$ in the circuit.
 $\therefore$ If we delete the edges not in the circuit, the graph will be like this:



Apparently nor a circuit. Contradict.

In conclusion, this graph does not have a Hamilton circuit